STConGeo: Robust Cross-View Geolocalization via Learnable Orientation Alignment and Straight-Through Estimation

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***Abstract*—In the task of cross-view image geo-localization (CVGL), the orientation alignment between ground panoramas and remote sensing images critically determines matching accuracy. While existing methods typically assume strict alignment, real-world panoramas often exhibit orientation uncertainty due to varying shooting angles and device orientations. This significantly degrades retrieval accuracy and matching stability. To address this challenge, this paper proposes a novel geo-localization model, namely STConGeo. Specifically, it introduces a Spatial Transformer Network (STN) to automatically learn and adaptively correct panoramic orientation offsets during end-to-end training. Furthermore, a Straight-Through Estimator (STE) is incorporated to overcome gradient propagation blockages caused by discrete modulo operations during the cyclic panoramic shifts of STN. Extensive experiments on the CVUSA_360 and CVACT_360 datasets demonstrate that STConGeo significantly improves matching accuracy while ensuring training stability.**

Keywords—Cross-view Image Geo-localization; Orientation Misalignment; Spatial Transformer Network; Straight-Through Estimator

I. INTRODUCTION

Cross-view Image Geolocalization (CVGL) is an advanced computer vision technology for geolocation using multi-source images. Its objective is to retrieve a corresponding geo-tagged reference image for a given query image from a large-scale dataset, thereby inferring the query's geographic location. Typically, the query is a ground-level street view or panorama, while the reference is an aerial or satellite remote sensing image [1][2]. With the explosive growth of unmanned aerial vehicles, satellite imagery, and street-view data, CVGL demonstrates immense potential and technical value in autonomous driving, robotic visual navigation, and 3D reconstruction [3][4][5].

Currently, although cross-view matching methods based on deep neural networks have achieved remarkable progress, most mainstream studies and their dependent datasets, such as the original CVUSA and CVACT, are built upon an idealized assumption: ground panoramas and remote sensing images are strictly pre-aligned to true north [6][7]. However, this assumption rarely holds in real-world scenarios, where factors like random shooting angles introduce substantial orientation uncertainty. This misalignment severely challenges cross-view matching. Specifically, random orientation offsets cause drastic visual displacements of key spatial layouts, dramatically increasing cross-modal matching difficulty. Consequently, traditional networks relying on known orientations suffer from sharply degraded adaptability and unstable matching accuracy when confronting such uncertainty.

To mitigate matching failures induced by orientation misalignment, recent research has explored targeted orientation compensation and feature alignment strategies. Vo et al. [8] bolstered rotation invariance via data augmentation and auxiliary orientation regression. Zhai et al. [9] utilized semantic segmentation priors to predict ground semantic maps from aerial images, selecting optimal matching angles based on cross-entropy similarity. Additionally, Shi et al. [10] proposed Dynamic Similarity Matching to estimate relative orientation via feature correlation along the azimuth axis. Focusing on activation maps, Zhu et al. [11] integrated Grad-CAM with pixel angular distributions to estimate orientation offsets. More recently, representations like W2W-BEV [12] and ConGeo [13] have also shown positive progress in tackling scenarios with unknown orientations and restricted fields of view.

To address real-world orientation uncertainty and existing deficiencies in feature alignment and training stability [14], we propose a novel cross-view image geo-localization method, namely STConGeo. The core breakthroughs include:

- Learnable Orientation Alignment via Spatial Transformer Network (STN): By embedding a STN into our architecture, the model learns to predict and correct the orientation offset of ground panoramas in an end-to-end manner, removing the need for prior orientation calibration.
- Gradient Flow Optimization via Straight-Through Estimator (STE): To handle the zero-gradient issue caused by the modulo-based cyclic padding during image shifting, we incorporate a STE. This mechanism preserves exact spatial transformations in the forward pass while ensuring stable gradient backpropagation via identity approximation, significantly enhancing model convergence [15].

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II. Proposed Method

A. Overview

Built upon ConGeo, the STConGeo architecture, as illustrated in Fig. 1, comprises three key stages. First, remote sensing images undergo polar transformation to align with panoramic coordinates, supplemented by augmentations. Second, an STN adaptively corrects panoramic orientation offsets, while an STE bypasses modulo-induced gradient blockages to ensure stable end-to-end training. Finally, a ConvNeXt extractor captures deep features from both modalities for precise localization via cosine similarity.

B. STN Module

STN is a differentiable module that dynamically generates geometric transformations based on input characteristics, rectifying data into a canonical form to facilitate feature extraction. Its lightweight, modular design allows seamless integration into CNNs with negligible computational overhead, and cascading STNs further captures complex variations. Within the STConGeo framework, the STN specifically learns horizontal orientation offsets, dynamically translating ground panoramas to align with the visual orientation of images. It includes three components: a localization network, a grid generator and a sampler.

Localization Network: This lightweight CNN regresses spatial transformation parameters using five convolutional layers and an adaptive global average pooling layer. This architecture captures macroscopic scene layouts and maps extracted features via a fully connected layer to a single horizontal translation parameter t_x. Optimized by the loss function, t_x adaptively aligns the ground panorama's orientation with the remote sensing imagery.

Grid Generator: Upon receiving the predicted translation parameter t_x, the grid generator establishes a precise coordinate mapping from the target image space back to the input space. It encodes t_x into a 2D affine transformation matrix θ, constrained exclusively for horizontal translation. Using θ, it executes an inverse spatial mapping by projecting each pixel coordinate from the target grid backward into the input panorama. This yields a parameterized sampling grid, ensuring the input pixels are correctly sampled to form the aligned target image.

Sampler: Guided by the generated grid, the sampler resamples the input panorama using bilinear interpolation to prevent floating-point distortion [16]. As formulated in (1), it approximates sub-pixel values via a spatially-weighted average of four adjacent input pixels. $I_{out}(x', y')$, $I_{in}(x_i, y_j)$, and $w_{i,j}$ denote the transform-8ed target pixel, original neighboring pixels, and spatial interpolation weights, respectively. This explicitly remaps the image into the target coordinate system. By horizontally translating the panorama, the sampler aligns its orientation with remote sensing imagery, enhancing spatial consistency and facilitating subsequent cross-view geo-localization tasks.

$$I_{out}(x', y') = \sum_{i,j} w_{i,j} \cdot I_{in}(x_i, y_j) \tag{1}$$

C. STE Mechanism

To prevent information loss caused by zero-padding during the STN's horizontal translation, a modulo operation is applied to the panoramic coordinates. However, this discrete function yields zero gradients, effectively blocking backpropagation. To resolve this, STConGeo employs a STE. By logically decoupling the forward and backward passes, the STE approximates these gradients, ensuring continuous and stable end-to-end optimization.

Forward Propagation: The network strictly preserves and executes the precise modulo operation to ensure the accurate physical mapping of the panoramic image during horizontal translation. Let the original input coordinate be denoted as X_r, the mapped coordinate as X_q, and m be the modulo base corresponding to the image width. The forward computation process is formulated as (2).

$$Forward: X_q = X_r \bmod m \tag{2}$$

Back Propagation: During backpropagation to the modulo node, the STE employs an identity function to approximate the gradient by setting it to 1, thereby enabling the lossless transmission of the error signal to the preceding layers. Its backpropagation formula is presented as (3).

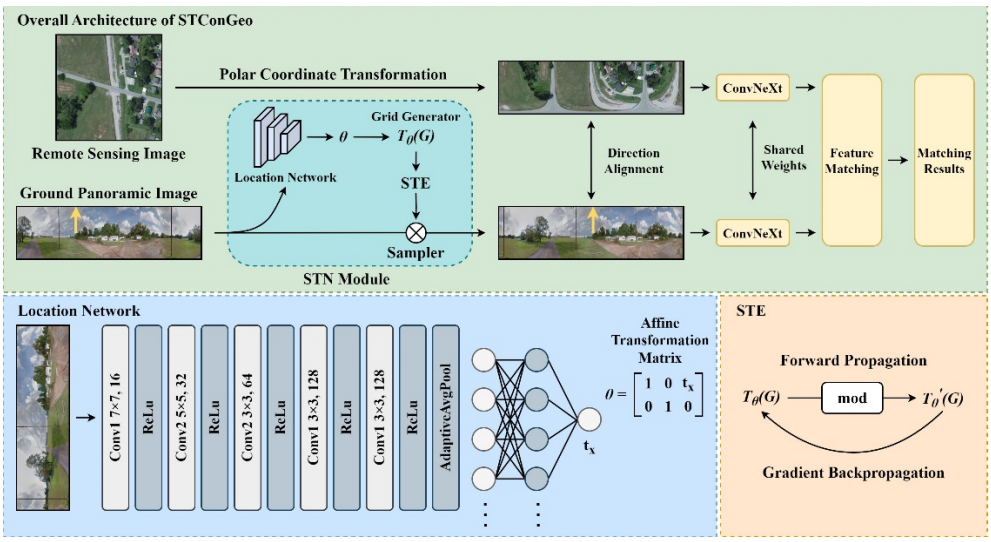

Fig. 1. Overall architecture of STConGeo. Remote sensing images undergo polar transformation, and ground panoramas are adaptively aligned by a STN module. A STE bypasses the non-differentiable modulo operation to ensure end-to-end training. Finally, a shared ConvNeXt extracts features for matching.

$$Backward: \frac{\partial Loss}{\partial X_r} \overset{STE}{=} \frac{\partial Loss}{\partial X_q} \tag{3}$$

By introducing STE, STConGeo circumvents the gradient vanishing trap, ensuring continuous error signal flow. This unlocks the STN's capacity for complex spatial transformations, granting the model robust end-to-end training stability and rapid convergence.

III. Experiments

A. Dataset

To evaluate the effectiveness of STConGeo, we used the public CVUSA [6] and CVACT [7] datasets, in which CVUSA covers diverse US areas, whereas CVACT samples densely from Canberra, Australia. Each dataset contains 35,532 training and 8,884 testing pairs. Both datasets were applied random horizontal panoramic offsets to simulate unknown real-world orientations. The modified datasets were named CVUSA_360 and CVACT_360, respectively.

B. Evaluation Metrics

In this experiment, Recall@K, abbreviated as R@K, is employed as the core evaluation metric. Its calculation formula is presented as (4) [6][7][10][17]. R@K measures the proportion of queries where the ground-truth image appears in the top-K retrieved results. In this study, K is set to 1, 5, 10, and 1%. Specifically, R@1 serves as the most critical metric for precise localization, whereas R@5, R@10, and R@1% evaluate broader recall capability and matching stability.

$$R@K = \frac{1}{n}\sum_{i=1}^{n} S_{i,k} \tag{4}$$

C. Implementation Details

We selected six cross-view geo-localization baselines: SAIG-D, TransGeo, Sample4Geo, GeoDTR, GeoDTR+, and ConGeo [13][17][18][19][20][21]. All models were trained for 100 epochs on both datasets with a batch size of 16. We employed the AdamW optimizer, decaying the initial learning rate of 0.001 to 1e-6 via cosine annealing. To ensure fairness, all experiments were conducted on a single NVIDIA GeForce RTX 3080Ti (12GB) GPU.

Following the pipeline of ConGeo, images underwent polar transformations alongside standard geometric and color augmentations. STConGeo was optimized by a combination of single-view contrastive, cross-view contrastive, and weighted soft-margin triplet losses, effectively regulating positive and negative sample distances within the joint feature space.

D. Experimental Results and Analysis

The results and analysis cover quantitative retrieval accuracy, qualitative visualizations, and ablation studies of core mechanisms.

Quantitative Retrieval Accuracy: As shown in TABLE I. , integrating the STN module significantly improves the performance of STConGeo on both orientation-misaligned datasets. On the diverse CVUSA_360 dataset, STConGeo achieves a R@1 of 89.97%, outperforming the best baseline, ConGeo, by 1.74%. Its R@5 and R@10 reach 98.58% and 99.34%, respectively. On the densely sampled CVACT_360, the R@1 of STConGeo increases from 75.20% to 76.38%, and the R@5 improves from 85.93% to 86.40%. These results demonstrate that the STN module effectively mitigates orientation uncertainty, enhancing localization accuracy and robustness across diverse distributions.

Qualitative Visualization: Fig. 2 illustrates the efficacy of STN in correcting orientation offsets. Examples display the query, STN-adjusted image, and Top-4 results. Yellow arrows indicate due north; orange borders highlight correct matches. Comparisons reveal that the module adaptively predicts offsets and horizontally translates panoramas toward due north. This correction minimizes orientation discrepancies, enabling precise visual and directional alignment in retrieved reference images.

TABLE I. Comparison of Recall Rates of Different Models and STConGeo on the CVUSA_360 and CVACT_360 Datasets

Model	CVUSA_360				CVACT_360			
	R@1	*R@5*	*R@10*	*R@1%*	*R@1*	*R@5*	*R@10*	*R@1%*
SAIG-D[18]	7.61%	20.09%	28.69%	63.39%	6.77%	19.45%	24.03%	54.35%
TransGeo[19]	12.70%	23.77%	29.96%	55.85%	11.90%	18.39%	21.90%	39.91%
Sample4Geo[20]	53.49%	71.09%	76.25%	88.39%	52.84%	57.72%	60.17%	71.17%
GeoDTR[17]	43.02%	70.76%	80.64%	97.29%	44.77%	68.92%	76.90%	92.92%
GeoDTR+[21]	77.21%	92.49%	95.88%	99.47%	70.08%	86.24%	**90.36%**	**96.36%**
ConGeo[13]	88.23%	94.78%	96.07%	98.24%	75.20%	85.93%	88.41%	94.17%
STConGeo	**89.97%**	**98.58%**	**99.34%**	**99.98%**	**76.38%**	**86.40%**	<u>88.69%</u>	<u>94.20%</u>

[a.] The best results are highlighted in bold and the second-best are underlined.

TABLE II. Impact of the STE Mechanism on the Recall Rate of the STConGeo Model

Model	CVUSA_360				CVACT_360			
	R@1	*R@5*	*R@10*	*R@1%*	*R@1*	*R@5*	*R@10*	*R@1%*
STConGeo (Without STE)	87.33%	94.35%	95.54%	98.19%	75.81%	85.48%	87.76%	93.40%
STConGeo (With STE)	89.97%	98.58%	99.34%	99.98%	76.38%	86.40%	88.69%	94.20%

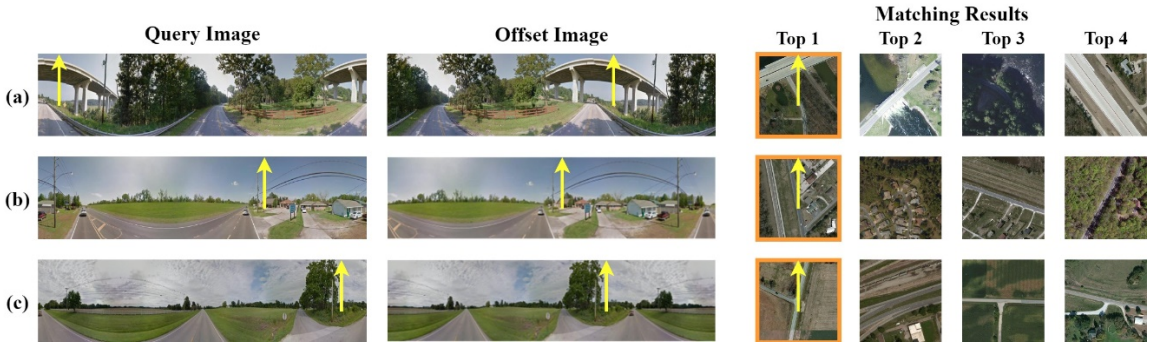

Fig. 2. Visualization results of STConGeo. For each set of examples, the images from left to right are the query image, the orientation-corrected image by the STN, and the top-4 retrieved results. Yellow arrows denote the true north, and orange bounding boxes indicate the correctly matched remote sensing images.

Ablation Study on STE: As shown in TABLE II. , STConGeo with STE significantly outperforms its ablated counterpart on CVUSA_360 in R@1 accuracy. This highlights the necessity of continuous gradient flow, which demonstrates that STE successfully resolves translation-induced gradient blockages, substantially enhancing end-to-end training robustness and efficiency.

IV. Conclusion

To address real-world orientation misalignment, this paper proposes the STConGeo geo-localization method. The main conclusions are drawn as follows: (1)By adaptively predicting and correcting panoramic orientation offsets, the embedded STN breaks the reliance on orientation priors, effectively overcoming spatial geometric misalignment. (2)The STE resolves the conflict between the discrete modulo operation and network differentiability. By decoupling precise forward mapping from identity backward approximation, it unblocks gradient propagation, guaranteeing stable end-to-end optimization. (3)Collaboratively optimizing orientation compensation and feature retrieval within an end-to-end architecture significantly boosts resistance to real-world geometric distortions, providing a robust paradigm for visual localization in unconstrained environments.

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